

Problem Statement

Heart Disease Analysis

Background

Cardiovascular disease remains one of the leading causes of death worldwide. Early identification of at-risk individuals can significantly improve treatment outcomes, but manual review of patient records is time-consuming and prone to oversight, especially as the volume of clinical data grows.

Problem Statement

Healthcare providers and analysts currently lack a fast, data-driven way to identify patterns and risk factors associated with heart disease from routinely collected clinical attributes (such as age, cholesterol, resting blood pressure, chest pain type, and maximum heart rate). Without a structured analytical approach, key risk signals in the data may go unnoticed, delaying preventive intervention.

Who is affected

- Cardiologists and clinicians who need quick, evidence-based risk indicators
- Healthcare analysts responsible for reporting on patient risk trends
- Patients, who benefit indirectly from earlier and more accurate risk detection

Goal

To analyze a heart disease clinical dataset in order to uncover the key factors most strongly associated with the presence of heart disease, and to present these findings through clear visualizations and, where applicable, a predictive risk model — enabling faster, more informed decisions.

Success Criteria

- A cleaned, analysis-ready version of the dataset
- Clear identification of top risk factors correlated with heart disease
- Visual dashboard summarizing key findings
- A final report communicating insights to a non-technical audience